MACHINE LEARNING WITH PYTHON

Log Sheet

Sl. No	Date	Topic Covered	Signature
1		Introduction to Machine Learning	
2		Artificial Intelligence vs Machine Learning vs Deep Learning	
3		Types of Machine Learning	
4		Machine Learning Workflow	
5		Python for Machine Learning (NumPy, Pandas, Matplotlib)	
6		Setting Up ML Environment (Colab, Jupyter)	
7		Understanding Data & Dataset Structure	
8		Basic Statistics for Machine Learning	
9		Introduction to Data Visualization	
10		Case Study & Foundations Review	
11		Introduction to Data Processing	
12		Handling Missing Values	
13		Data Cleansing Techniques	
14		Feature Scaling (Normalization & Standardization)	
15		Encoding Categorical Data	
16		Handling Imbalanced Datasets (SMOTE, Near Miss)	
17		Train-Test Split & Avoiding Data Leakage	
18		Cross-Validation Techniques	
19		Feature Engineering	
20		Data Preparation Review & Practical Implementation	
21		Introduction to Supervised Learning	
22		Linear Regression	
23		Multiple & Polynomial Regression	
24		Logistic Regression	
25		Naive Bayes Classifier	
26		K-Nearest Neighbors (KNN)	
27		Support Vector Machine (SVM)	
28		Decision Trees	
29		Random Forest & Ensemble Learning	
30		Model Evaluation & Supervised Learning Review	